



Efficacy and safety of vaccines to prevent respiratory syncytial virus infection in infants and older adults: A systematic review and meta-analysis[☆]

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ABSTRACT

Objectives: To determine the efficacy and safety of respiratory syncytial virus (RSV) vaccines in infants and older adults.

Methods: We performed a systematic review and meta-analysis of randomized control trials that evaluated the efficacy of maternal RSV immunization against infections in infants, as well as the efficacy of RSV vaccines in older adults. The primary outcome was the vaccine efficacy against RSV-related lower respiratory tract disease (LRTD). Grading of Recommendations Assessment, Development and Evaluation criteria was used to evaluate the level of evidence.

Results: Ten trials were included in the review. For maternal vaccination, the RSV vaccine showed favourable efficacy against RSV-related LRTD (vaccine efficacy 57.3%, 95% confidence interval [CI] 31.3–73.5; low certainty) and RSV-related severe LRTD (vaccine efficacy 81.9%, 95% CI 56.8–92.4; moderate certainty) in infants within 90 days after birth. For older adults, Meta-analysis showed that RSV vaccines could also reduce the risk of RSV-related LRTD (vaccine efficacy 78.3%, 95% CI 65.6–86.3; moderate certainty) and RSV-related severe LRTD (vaccine efficacy 86.5%, 95% CI 68.3–94.3; moderate certainty). There was no significant difference in serious adverse events between RSV vaccines and placebo.

Conclusion: RSV vaccines have the potential to offer protection against RSV disease in both infants and older adults, without apparent safety concerns.

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Introduction

Respiratory syncytial virus (RSV) infection is a widespread cause of lower respiratory tract disease (LRTD) worldwide, posing a significant risk of severe illness among infants and older adults [1]. RSV causes an estimated 33.1 million acute lower respiratory tract

infections, 3.2 million hospitalizations and 101.4 thousand deaths in children under 5 years of age annually worldwide [2]. Immunizing pregnant women or healthy term-born infants during their first winter season could significantly reduce the healthcare burden associated with RSV [3]. RSV is a leading cause of LRTD in older adults and individuals with pre-existing chronic cardiac or pulmonary conditions [4]. In industrialized countries, there were an estimated 5.2 million cases of acute respiratory infection (ARI), 470 thousand hospitalizations and 33 thousand in-hospital deaths among adults aged 60 years or older attributed to RSV infection in 2019 [5]. Nevertheless, the burden of RSV in children and older adults is likely underestimated, and our understanding of the disease's impact has only recently begun to address significant knowledge gaps [6].

[☆] This study was registered on PROSPERO (CRD42023464623) https://www.crd.york.ac.uk/PROSPERO/display_record.php?RecordID=464623.

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Efforts to develop a vaccine for RSV have indeed remained highly active despite the initial setback in the 1960s with the formalin-inactivated RSV vaccine [1]. Currently, there are numerous RSV prevention candidates in clinical development utilizing six different approaches: recombinant vector, subunit, particle-based, live attenuated, chimeric and nucleic acid vaccines [6]. These diverse approaches reflect ongoing research and commitment to finding an effective and safe RSV vaccine. In recent years, significant progress has been made in the development of RSV vaccines. Arexvy (RSVPreF3) became the first RSV vaccine to receive approval from the US Food and Drug Administration (FDA) in May 2023 [7]. It is specifically approved for the prevention of LRTD caused by RSV in individuals aged 60 years and older. Additionally, in August 2023, the FDA approved Abrysvo (RSVpreF), which is the first vaccine authorized for use in pregnant women to prevent LRTD and severe LRTD caused by RSV in infants from birth through 6 months of age [8]. However, it's worth noting that not all RSV vaccine candidates have shown efficacy in clinical trials. Two trials evaluating RSV vaccines in adults aged 60 years or older, one based on RSV postfusion F protein and the other utilizing a nanoparticle-based RSV-F vaccine, did not demonstrate protection against RSV-related illness [9,10]. The synthesis of evidence regarding RSV vaccines for the prevention of infections in infants and older adults is indeed urgent and necessary. It is crucial to assess the potential benefits and risks of RSV vaccines in these vulnerable populations. Therefore, we performed a systematic review and meta-analysis of phase 2/3 randomized controlled trials (RCTs) to determine the efficacy and safety of RSV vaccines in infants and older adults.

Methods

Data sources and searches

We conducted this meta-analysis according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [11], and the protocol was registered on PROSPERO. We searched for literature published on PubMed, Embase, Cochrane Library and grey literature from the ClinicalTrials.gov website until 1 September 2023. If a trial is registered on ClinicalTrials.gov but does not have any available results, it is common practice to search for additional data related to the study on other platforms, such as searching on Google. Keywords including 'respiratory syncytial virus', 'RSV', 'vaccine', 'vaccination', 'immunization' and 'randomized control trial' were used to search, and the detailed search strategy was shown in the Supplementary material. Additionally, we identified references by searching the reference lists of included studies and relevant reviews. Considering that the researches on RSV vaccines were updated quickly, we researched the literature before submission on December 15, 2023.

Study selection

We included phase 2/3 RCTs evaluating the efficacy of RSV vaccines in target populations (maternal vaccination to protect against infections in infants, and adults aged 60 or older), irrespective of RSV vaccine type, control group or length of follow-up. We excluded phase 1/2 trials that did not report vaccine efficacy. Additionally, study protocols, editorials, comments, reviews, conference abstracts, ongoing trials and studies without sufficient data for analysis were also excluded from our review. The primary outcome was the vaccine efficacy against RSV-related LRTD depending on the definition of trials. Secondary outcomes included vaccine efficacy against RSV-related ARI, RSV-related severe LRTD and hospitalization for RSV-related LRTD. Safety outcomes, such as any adverse events (AEs) and serious AEs (SAEs), were also considered

as part of the secondary outcomes. AEs and SAEs were defined according to the definitions of each included study.

Data extraction and quality assessment

Two authors reviewed titles and abstracts independently to identify eligible studies that met pre-specified inclusion criteria and extracted data. When consensus was lacking, a third reviewer was consulted. Study characteristics (e.g. journal and year of publication, study design, sample size and length of follow-up), intervention and comparator characteristics (e.g. type of vaccine, manufacturer and status), participant characteristics (e.g. age and sex), and outcomes were extracted. RSV-related LRTD with at least 3 symptoms was regarded as RSV-related severe LRTD. The risk of bias of RCTs was assessed by the Cochrane Collaboration's tool (RoB 2) [12], which includes the following domains: randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. The quality of evidence for each outcome was evaluated with the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework [13].

Data synthesis and analysis

Initially, we conducted a narrative descriptive synthesis to evaluate the vaccine efficacy and safety of the included RCTs. Based on their developmental status, the vaccines were categorized into three groups: approved vaccines, promising vaccines in the approval process, and failed or discontinued vaccines. Approved vaccines are those that have received market approval from regulatory authorities. Promising vaccines are those showing favourable results in the approval process. Discontinued vaccines are those that manufacturers have announced will no longer be developed or marketed. Only approved and promising vaccines were included in our quantitative analysis to evaluate the potential benefits and risks of RSV vaccines that are more likely to be available for use in the foreseeable future. The two target populations, infants and older adults were analysed separately. We conducted DerSimonian and Laird random-effects meta-analysis when data were available for at least two studies [14]. Data were analysed as relative risk or absolute risk with 95% confidence intervals (CIs). Vaccine efficacy was defined as $(1 - \text{relative risk}) \times 100\%$. Statistical heterogeneity between the studies was assessed with the χ^2 test and the I^2 statistics. I^2 values of 25%, 50% and 75% have been suggested to be indicators of low, moderate and high heterogeneity, respectively [15]. All the analysis were performed with Stata 17.

Results

Characteristics of included studies

This systematic literature search initially identified 1940 records, after excluding duplicates and irrelevant records, 48 studies were evaluated in full text for eligibility (Figure 1). Finally, 10 multicentre RCTs (12,400 pregnant women with 12,102 infants and 115,653 older adults) were included in the present study [10,16–24]. Among the included studies, eight [10,16–18,21–24] were published, and two [19,20] were unpublished data. Of the 10 trials, three [17,18,22] focused on vaccination during pregnancy and evaluated vaccine efficacy in infants, while the remaining seven trials assessed vaccine efficacy in older adults. A total of six RSV vaccines in 10 trials met the inclusion criteria for this review. These included Pfizer RSVpreF (3 trials [17,22,23], approved for pregnant women and older adults), GlaxoSmithKline RSVPreF3 OA (1 trial [21], approved for older adults), Moderna mRNA-1345 (1 trial [24], a promising vaccine for older adults), Janssen Ad26.RSV.preF

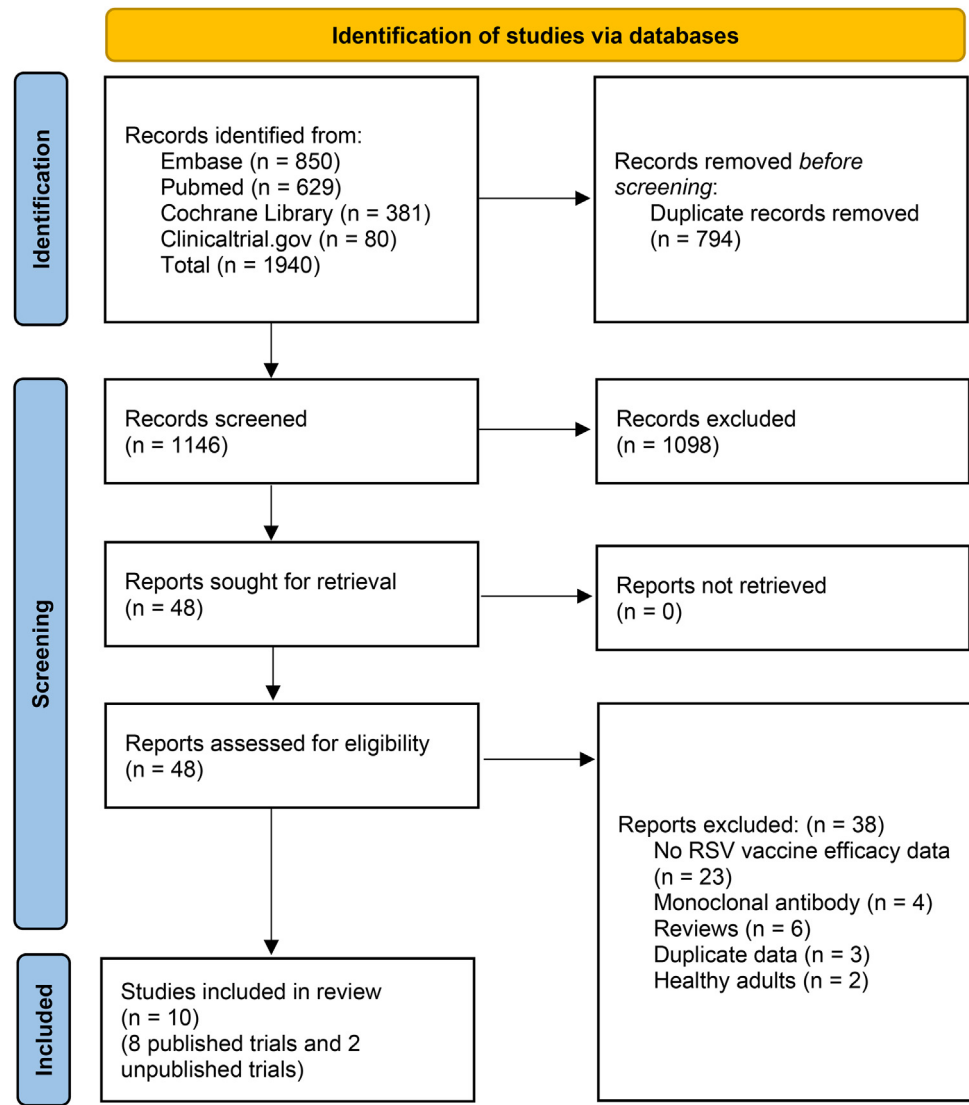


Figure 1. Study selection.

(1 trial [16], discontinued for older adults), Novavax RSV F vaccine (3 trials [18–20], failure for pregnant women and older adults) and MedImmune MEDI7510 (1 trial [10], failure for older adults). Finally, two trials [17,22] of pregnant women ($n = 7736$) with their infants ($n = 7528$) and three trials [21,23,24] of older adults ($n = 94,791$) were included to evaluate the potential benefits and risks of RSV vaccines. Characteristics of individual studies are summarised in Table 1. Most examined items were assessed as low risk. The detailed risk of bias assessment is available in Table S1. The quality of evidence for the main outcomes was rated following the GRADE framework (Table S2).

Vaccine efficacy of maternal immunization to protect against infections in infants

Three trials [17,18,22] reported the efficacy of RSV vaccination during pregnancy to protect against infections in infants. Two trials [17,22] were conducted on RSVpreF, and one trial [18] was conducted on RSV F vaccine. The relative risk for RSV-related LRTD, RSV-related severe LRTD, and hospitalization for RSV-related LRTD appeared to be lower for RSVpreF compared to RSV F vaccine, both at 90 and 180 days (Figure S1). Noted that the reported vaccine efficacy of one phase 2b trial of RSVpreF in infants was

based on exploratory, post hoc analyses [22], which might cause bias. Therefore, only one trial [17] was included to evaluate the potential benefits of the RSV vaccine (RSVpreF). The RSV vaccine showed favourable efficacy against RSV-related LRTD (vaccine efficacy 57.3%, 95% CI 31.3–73.5; absolute risk, 92 fewer per 10,000 patients, 95% CI 42–142; low certainty), RSV-related severe LRTD (vaccine efficacy 81.9%, 95% CI 56.8–92.4; absolute risk, 78 fewer per 10,000 patients, 95% CI 43–113; moderate certainty) and hospitalization for RSV-related LRTD (vaccine efficacy 67.9%, 95% CI 34.6–84.2; absolute risk, 60 fewer per 10,000 patients, 95% CI 25–96; low certainty) within 90 days after birth. The RSV vaccine efficacy was slightly lower at 180 days after birth compared to the efficacy observed at 90 days, as shown in Table 2.

Vaccine efficacy in older adults

There were seven trials [10,16,19–21,23,24] that reported the efficacy of RSV vaccines in older adults (Figure S2). These trials evaluated different RSV vaccines, including two trials [19,20] for RSV F vaccine, one [23] for RSVpreF, one [21] for RSVpreF3 OA, one [16] for Ad26.RSV.preF, one [10] for MEDI7510 and one [24] for mRNA-1345. Three trials of two vaccines (RSV F vaccine and MEDI7510) showed ineffective for older adults [10,19,20]

Table 1
Summary of included randomised controlled trials.

Study	Participant characteristics	Study design and registration	Vaccines	Control	No of participants	Length of follow-up (months)	Status
Kampmann [17] N Engl J Med	Pregnant women; age range 49 years or younger; gestation 24–36 weeks	Multicentre (18 countries), phase 3, NCT04424316	RSVpreF (Pfizer), single dose of 120 µg	Placebo	7358	24	Approved
Simões [22] N Engl J Med	Pregnant women; age range 18–49; Gestation 24–36 weeks	Multicentre (4 countries), phase 2b, NCT04032093	RSVpreF (Pfizer), single dose of 120 or 240 µg, with or without aluminium hydroxide	Placebo	406	6	Approved
Madhi [18] N Engl J Med	Pregnant women; age range 14–40; gestation 28–36 weeks	Multicentre (11 countries), phase 3, NCT02624947	RSV F vaccine (Novavax), single dose of 120 µg adsorbed to 0.4 mg of aluminium	Placebo	4636	12	Failure
Walsh [23] N Engl J Med	Older Adults; age ≥60 years; male 50.8%	Multicentre (7 countries), phase 3, NCT05035212	RSVpreF (Pfizer), single dose of 120 µg	Placebo	34,284	6	Approved
Papi [21] N Engl J Med	Older Adults; age ≥60 years; male 48.3%	Multicentre (17 countries), phase 3, NCT04886596	RSVPreF3 OA (GlaxoSmithKline), single dose of 120 µg, AS01-adjuvanted	Placebo	24,966	6	Approved
Wilson [24] N Engl J Med	Older Adults; age ≥60 years; male 50.9%	Multicentre (22 countries), phase 2/3, NCT05127434	mRNA-1345 (Moderna), single dose of 50 µg	Placebo	35,541	12	Promising
Falsey [16] N Engl J Med	Older Adults; age ≥65 years; male 42.3%	Multicentre (United States), phase 2b, NCT03982199	Ad26.RSV.preF (Janssen), single dose of 150 µg	Placebo	5782	6	Discontinued
Falloon [10] J Infect Dis	Older Adults; age ≥60 years; male 58.9%	Multicentre (7 countries), phase 2b, NCT02508194	MEDI7510 (MedImmune), single dose of 125 µg, adjuvanted inactivated influenza vaccine	Placebo	1900	12	Failure
Novavax_phase3 [19] 2016/Clinical-Trials.gov	Older Adults; age ≥60 years; male NR	Multicentre (United States), phase 3, NCT02608502	RSV F vaccine (Novavax), single dose of 135 µg	Placebo	11,850	6	Failure
Novavax_phase2 [20] 2016/Clinical-Trials.gov	Older Adults; age ≥60 years; male NR	Multicentre (United States), phase 2, NCT02593071	RSV F vaccine (Novavax), single dose of 135 µg	Placebo	1330	6	Failure

(Figure S2). Despite favourable efficacy against RSV-LRTD observed in the phase 2b trial [16], the manufacturer of Ad26.RSV.preF, Johnson & Johnson, announced the discontinuation of its phase 3 EV-ERGREEN study (NCT04908683) [25]. Therefore, only three trials [21,23,24] were included to evaluate the potential benefits of RSV vaccines (Figure 2). Meta-analysis showed that RSV vaccines could reduce the risk of RSV-related LRTD (vaccine efficacy 78.3%, 95% CI 65.6% to 86.3%; absolute risk, 22 fewer per 10,000 patients, 95% CI 13–31; moderate certainty), RSV-related ARI (vaccine efficacy 68.0%, 95% CI 58.5–75.3; absolute risk, 35 fewer per 10,000 patients, 95% CI 19–51; moderate certainty) and RSV-related severe LRTD (vaccine efficacy 86.5%, 95% CI 68.3–94.3; absolute risk, 9 fewer per 10,000 patients, 95% CI 6–12; moderate certainty). Table 2 presents the vaccine efficacy of three different vaccines.

Safety of RSV vaccines

Three trials [17,18,22] involving two vaccines, the RSV F vaccine and RSVpreF, reported any AEs and SAEs in both maternal and infant participants (Figure S3). The results indicated that the RSV F vaccine was associated with a higher risk of AEs among maternal participants (relative risk 1.08, 95% CI 1.04–1.11) and the RSVpreF showed a higher risk of AEs in infant participants (2 trials; relative risk 1.08, 95% CI 1.01–1.14; moderate certainty; Figure S4). However, there was no significant difference in SAEs between the RSV vaccines and the placebo, both in maternal and infant participants. Five trials [10,16,21,23,24] involving five different vaccines reported AEs and SAEs in older adults (Figure S3). Among the RSV vaccines, except for RSVPreF3 OA and mRNA-1345, there was no significant difference in safety outcomes between the vaccines and placebo in

older adults. However, both RSVPreF3 OA (relative risk 1.85, 95% CI 1.77–1.94) and mRNA-1345 (relative risk 1.08, 95% CI 1.04–1.13) were associated with a higher risk of AEs. Meta-analysis of two approved vaccines and one promising vaccine indicated that RSV vaccines have no significant difference in AEs (relative risk 1.28, 95% CI 0.88–1.94; very low certainty) and SAEs (relative risk 1.03, 95% CI 0.95–1.12; moderate certainty) compared to placebo (Table 2 and Figure 3).

Discussion

This systematic review and meta-analysis included 10 RCTs involving 12,400 pregnant women, 12,102 infants and 115,653 older adults. The primary objective was to provide a comprehensive overview of the efficacy and safety of RSV vaccines. Among the six vaccines included, only one will be used for pregnant women to prevent RSV infection in infants, and three of them will likely be used for older adults. The findings revealed that the RSV vaccine (RSVpreF) demonstrated positive outcomes in infants. Specifically, per 10,000 participants, the vaccine was associated with a decrease of 92 cases of RSV-related LRTD, 78 cases of RSV-related severe LRTD, and 60 cases of hospitalization for RSV-related LRTD within 90 days after birth. Regarding older adults, the RSV vaccines also showed promising results. Per 10,000 older adults, the vaccine was associated with a decrease of 22 cases of RSV-related LRTD, 35 cases of RSV-related ARI and 9 cases of RSV-related severe LRTD during the first RSV season following vaccination. Furthermore, no significant difference in SAEs was observed compared to the placebo group.

Table 2
Vaccine efficacy and safety of RSV vaccines within reach.

Participants	Type of vaccines	Outcomes	Vaccine (n/N)	Placebo (n/N)	Vaccine efficacy or risk ratio (95% CI)	Risk difference (95% CI) (per 10,000 participants)	GRADE certainty
Infants	RSVpreF	RSV-related severe LRTD (90 days)	6/3495	33/3480	VE 81.9% (56.8–92.4)	78 fewer (43–113)	Moderate
		RSV-related LRTD (90 days)	24/3495	56/3480	VE 57.3% (31.3–73.5)	92 fewer (42–142)	Low
		RSV-LRTD hospitalization (90 days)	10/3495	31/3480	VE 67.9% (34.6–84.2)	60 fewer (25–96)	Low
		RSV-related severe LRTD (180 days)	19/3495	62/3480	VE 69.5% (49.1–81.7)	124 fewer (44–174)	Low
		RSV-related LRTD (180 days)	19/3495	44/3480	VE 51.5% (33.7–64.5)	173 fewer (100–246)	Low
		RSV-LRTD hospitalization (180 days)	57/3495	117/3480	VE 57.0% (26.5–74.8)	72 fewer (28–116)	Low
Older adults	All vaccines	RSV-related LRTD	27/46,344	128/46,318	VE 78.3% (65.6–86.3)	22 fewer (13–31)	Moderate
		RSV-related ARI	75/46,344	235/46,318	VE 68.0% (58.5–75.3)	35 fewer (19–51)	Moderate
		RSV-related severe LRTD	6/46,344	48/46,318	VE 86.5% (68.3–94.3)	9 fewer (6–12)	Moderate
	RSVpreF	RSV-related LRTD (≥ 2 symptoms)	11/16,306	33/16,308	VE 66.7% (34.1–83.1)	14 fewer (6–22)	Low
		RSV-related ARI	22/16,306	58/16,308	VE 62.1% (38.1–76.8)	22 fewer (11–33)	Low
		RSV-related severe LRTD (≥ 3 symptoms)	2/16,306	14/16,308	VE 85.7% (37.1–96.8)	7 fewer (3–12)	Low
	RSVPreF3 OA	RSV-related LRTD (≥ 2 symptoms)	7/12,466	40/12,494	VE 82.5% (60.8–92.1)	26 fewer (16–37)	Moderate
		RSV-related ARI	27/12,466	95/12,494	VE 71.5% (56.3–81.4)	54 fewer (37–72)	Moderate
		RSV-related severe LRTD	1/12,466	17/12,494	VE 94.1% (55.7–99.2)	13 fewer (6–20)	Low
	mRNA-1345	RSV-related LRTD (≥ 2 symptoms)	9/17,572	55/17,516	VE 83.7% (67.0–91.9)	26 fewer (17–35)	Moderate
		RSV-related ARI	26/17,572	82/17,516	VE 68.4% (50.9–79.7)	32 fewer (20–44)	Moderate
		RSV-related severe LRTD (≥ 3 symptoms)	3/17,572	17/17,516	VE 82.4% (40.0–94.8)	8 fewer (3–13)	Low
Pregnant women	RSVpreF	Any adverse events	596/4009	496/3754	RR 1.11 (0.89–1.37)	259 (–362 to 881)	Moderate
Infants	RSVpreF	Serious adverse events	165/4009	138/3754	RR 1.14 (0.91–1.42)	51 (–36 to 138)	Moderate
		Any adverse events	1464/3892	1258/3636	RR 1.08 (1.01–1.14)	266 (47–485)	Moderate
Older adults	All vaccines	Serious adverse events	622/3892	553/3636	RR 1.04 (0.58–1.21)	117 (–280 to 514)	Moderate
		Any adverse events	9285/47,416	701/47,247	RR 1.28 (0.88–1.94)	574 (–224 to 1372)	Very low
	RSVpreF	Serious adverse events	1020/47,416	986/47,247	RR 1.03 (0.95–1.12)	5 (–28 to 35)	Moderate
		Any adverse events	1544/17,215	1453/17,069	RR 1.05 (0.98–1.13)	46 (–14 to 105)	Moderate
	mRNA-1345	Serious adverse events	396/17,215	387/17,069	RR 1.01 (0.88–1.17)	3 (–28 to 35)	Moderate
		Any adverse events	3624/17,734	3331/17,679	RR 1.08 (1.04–1.13)	159 (76–242)	Moderate
	RSVPreF3 OA	Serious adverse events	102/17,734	93/17,679	RR 1.09 (0.83–1.45)	5 (–11 to 20)	Moderate
		Any adverse events	4117/12,467	2229/12,499	RR 1.85 (1.77–1.94)	1519 (1413–1625)	Moderate
		Serious adverse events	522/12,467	506/12,499	RR 1.03 (0.92–1.17)	14 (–35 to 63)	Moderate
		Any adverse events					

Definitions of safety outcomes: (1. RSVpreF for pregnant women and infants: adverse events reported within 1 month after injection. 2. RSVpreF for older adults: data on unsolicited adverse events in all the participants were collected from enrollment through 1 month after injection. 3. mRNA-1345 for older adults: unsolicited treatment-emergent adverse events within 28 days after injection. 4. RSVPreF3 OA for older adults: unsolicited adverse events were included up to 30 days after injection, serious adverse events were included up to 6 months after injection). Definitions of RSV-related LRTD and severe LRTD: (1. RSVpreF for pregnant women and infants: medically attended RSV-associated LRTD; medically attended severe RSV-associated LRTD. 2. RSVpreF for older adults: RSV-associated LRTD with at least two signs or symptoms lasting more than 1 day and RSV infection that was confirmed by RT-PCR within 7 days after the onset of signs or symptoms; RSV-associated LRTD with at least three signs or symptoms lasting more than 1 day and RSV infection that was confirmed by RT-PCR within 7 days after the onset of signs or symptoms. 3. mRNA-1345 for older adults: RSV-associated LRTD with ≥ 2 signs or symptoms; RSV-associated LRTD with ≥ 3 signs or symptoms. 4. RSVPreF3 OA for older adults: RSV-LRTD as confirmed by RT-PCR, LRTD was defined as at least two lower respiratory symptoms or signs (including at least one lower respiratory sign) or at least three lower respiratory symptoms lasting for at least 24 hours. Severe disease was determined on the basis of either of two case definitions: on the basis of clinical signs or investigator assessment or on the basis of receipt of supportive therapy).

ARI, acute respiratory infection; CI, confidence interval; GRADE, Grading of Recommendations, Assessment, Development and Evaluation; LRTD, lower respiratory tract disease; RR, relative risk; RSV, respiratory syncytial virus; VE, vaccine efficacy.

To our knowledge, our study is the first comprehensive systematic review and meta-analysis to characterize the efficacy and safety of RSV vaccines. A recent review provided an overview of RSV vaccines under clinical development, highlighting various target populations, antigens and trial outcomes [6]. The review indicated that most vaccines included in the analysis demonstrated

safety and good tolerability based on phase 1/2 trials. However, some crucial phase 3 trial results were not available in that review. Similar findings were reported in a previous review that specifically examined the immunogenicity and safety of RSV vaccines under development [26]. Differences in efficacy were observed between approved vaccines and those that did not meet the desired

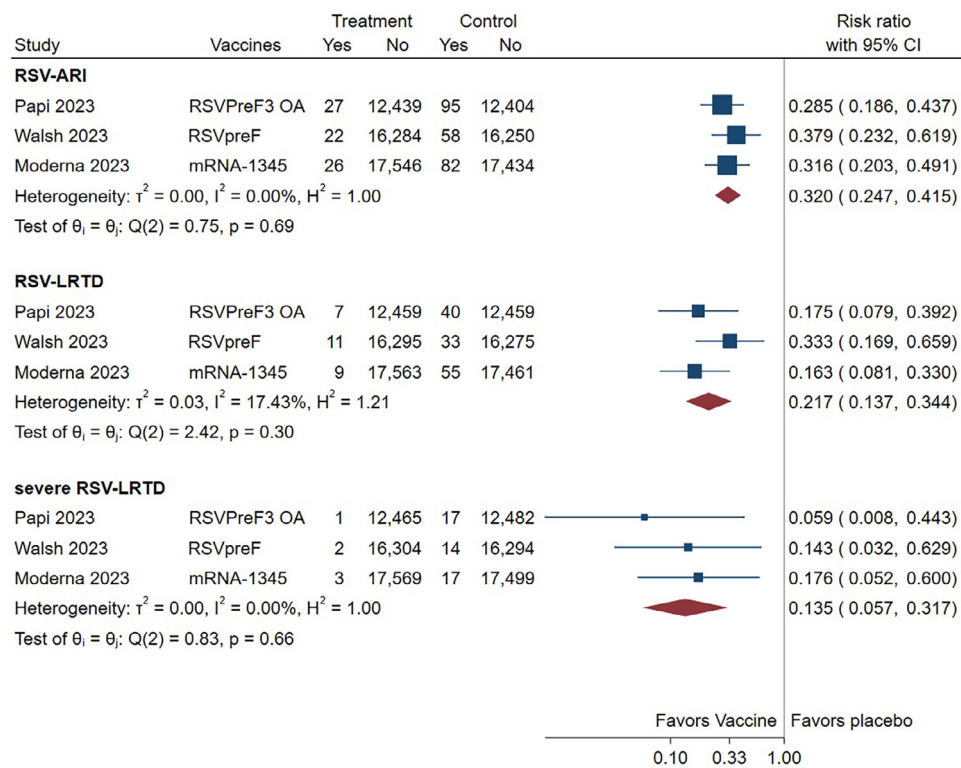


Figure 2. Meta-analysis of the efficacy of the approved and promising RSV vaccines in older adults.

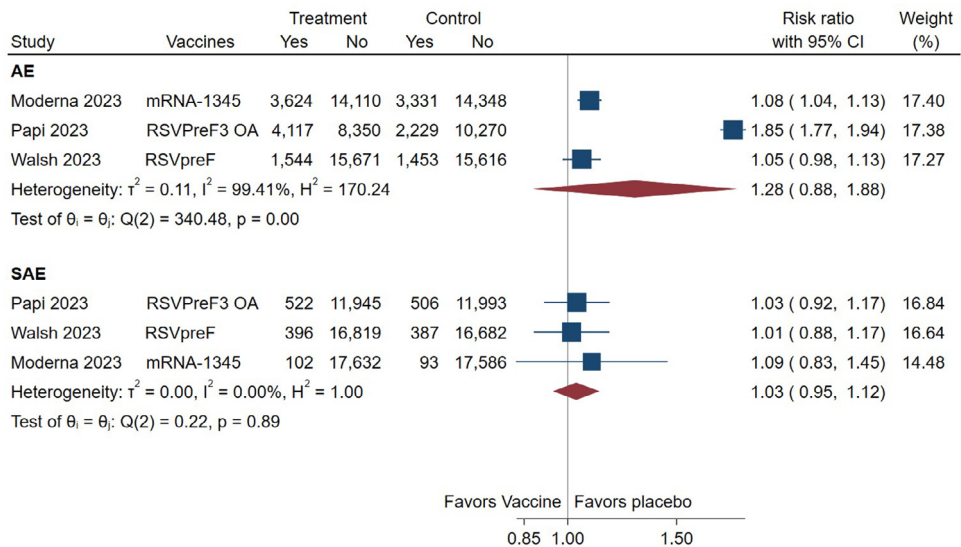


Figure 3. Meta-analysis of the safety of the approved and promising RSV vaccines in older adults.

outcomes, which could be attributed to variations in their development approaches. Two approved vaccines, RSVpreF and RSVPreF3 OA, are based on the RSV prefusion F protein. The promising vaccine, mRNA-1345, encoding the stabilized RSV prefusion F glycoprotein, utilizes a nucleic acid approach. However, the discontinued vaccine (Ad26.RSV.preF) employed a recombinant vector strategy. The two vaccines that failed, RSV F vaccine and MEDI7510, were an RSV F nanoparticle vaccine and a postfusion F protein-based vaccine, respectively.

While the primary outcome (RSV-associated medically significant LRTI) of the maternal Novavax RSV F vaccine administered during pregnancy did not meet the pre-specified criteria (lower bound of the 97.52% CI of $\geq 30\%$), moderate efficacy was demon-

strated for other endpoints related to RSV-associated and all-cause respiratory disease in infants. As a result, further research on the efficacy of the RSV F vaccine may be warranted in the future.

RSV-related LRTD poses a significant burden of illness and mortality among young infants worldwide, particularly in low- and middle-income countries [27]. Recently, nirsevimab, a monoclonal antibody, has been authorized for use in infants from birth during the RSV season based on three clinical trials [28-30]. Infants who received nirsevimab injection were protected from RSV-related LRTD for up to 150 days after the injection, demonstrating an efficacy of 74.5% (95% CI, 49.6-87.1) [28]. This efficacy appears to be slightly higher than that of RSVpreF vaccine, which demonstrated an efficacy of 52.5% (95% CI, 28.7-68.9) within 150 days af-

ter birth [22]. However, further evidence is needed to determine the true comparative effectiveness between the two interventions. Additionally, the affordability and cost-effectiveness of these interventions in low- and middle-income countries remain uncertain. Data from low-income countries where the vaccine is likely to have the greatest effect is limited for RSVpreF in infants [17]. Although the trial of RSV F vaccine was not powered to evaluate vaccine efficacy according to country income level, it appeared that the vaccine efficacy against RSV-related LRTD in infants was greater in low- or middle-income countries compared to high-income countries [18]. It is important to note that the efficacy and safety of RSV vaccines in preterm infants are not yet known. Additional studies are needed to determine if the variation in vaccine efficacy in infants between high-income and low- or middle-income countries is a consistent finding. Furthermore, the effectiveness of maternal RSV vaccination in preventing RSV-related LRTD in preterm infants should also be evaluated in the future.

This review has several strengths. First, the systematic review and meta-analysis followed the PRISMA guidelines and had a protocol registered in PROSPERO. We searched multiple databases comprehensively and included all available RCTs that reported the RSV vaccine efficacy. Second, the risk of bias for included trials was assessed using a valid methodological tool, and the quality of evidence for each outcome was evaluated using GRADE. Third, we have performed sufficient analysis in the review. A narrative descriptive synthesis was conducted for all included RSV vaccines to provide a comprehensive overview of their characteristics, efficacy and safety. However, only approved and promising vaccines were included to provide more precise quantitative data to inform decision-making and better understand the potential benefits and risks associated with these vaccines.

We also acknowledge some limitations. First, the limited number of enrolled trials is a result of the ongoing development of RSV vaccines. We could only include one trial for infants and three trials for older adults to evaluate the potential benefits of RSV vaccines. Second, it is important to note that a significant number of phase 1/2 trials only provided data on the safety and immunogenicity of the vaccines, which were not included in our analysis. This limitation may reduce the statistical power of our safety analysis since we did not have access to the complete safety data from these trials. Third, the effectiveness and safety of RSV vaccines in real-world settings remain unknown, and further studies are needed to address this gap. Fourth, the results of this meta-analysis may not apply to certain clinically relevant subgroups, such as preterm infants and frail older adults. Finally, due to the limited number of included trials, we were unable to perform subgroup analyses on vaccine efficacy against specific RSV subtypes (A and B) or evaluate vaccine efficacy in different age groups.

In conclusion, this systematic review and meta-analysis indicate that RSV vaccinations could offer protection against RSV disease in older adults as well as infants if administered during pregnancy, without apparent safety concerns. However, further evidence from real-world settings is necessary to strengthen these findings and evaluate the vaccine effectiveness in broader populations. Additionally, assessing the cost-effectiveness of RSV vaccines compared to monoclonal antibodies is crucial, considering factors such as efficacy, safety, duration of protection and overall economic impact. This information will be valuable for decision-making and resource allocation in RSV prevention strategies.

Declarations of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Author contributions

FS and BZ conceived the study, participated in its design and coordination and critically revised the manuscript. BZ had full access to all the data collection, analysis and interpretation and drafted the manuscript. BZ, XL, QY, JW and QR contributed to the process of data collection and data analyses as study investigators. All authors approved the final manuscript. FS and BZ are the guarantors. All authors had full access to all the data in the study, and the corresponding authors had final responsibility for the decision to submit for publication. The corresponding author (BZ) attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijid.2024.107118.

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